# Skin Lesion Classification Using Advanced Deep Learning

## 1. Introduction

Skin cancer is one of the most prevalent forms of cancer worldwide, where early detection plays a critical role in improving patient survival rates. Traditional diagnostic approaches rely heavily on dermatologists and may suffer from subjectivity and inter-observer variability.

With the advancement of artificial intelligence, deep learning—particularly Convolutional Neural Networks (CNNs)—has demonstrated strong potential in medical image analysis. These models can automatically learn complex visual patterns and assist in diagnostic decision-making.

This project develops a deep learning-based classification system using the HAM10000 dataset. Unlike a basic implementation, this work explicitly focuses on **handling class imbalance, improving model architecture, and applying rigorous evaluation techniques**, which are essential for reliable real-world medical applications.

## 2. Objectives

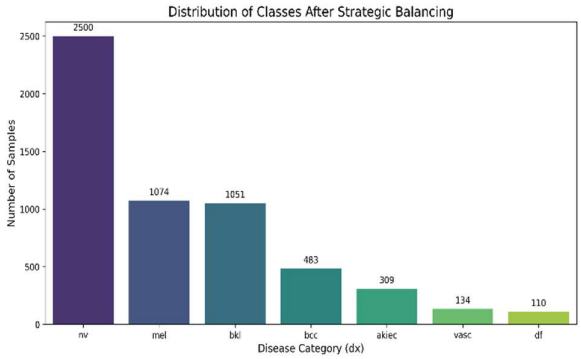
- Develop a robust multi-class classification model for skin lesions
- Address severe class imbalance in the dataset
- Apply advanced techniques such as transfer learning and fine-tuning
- Evaluate performance using appropriate metrics beyond accuracy
- Critically analyze model behavior and limitations

## 3. Dataset Description and Analysis

The HAM10000 dataset consists of **10,015 dermatoscopic images** categorized into seven diagnostic classes.

A major challenge in this dataset is **extreme class imbalance**, where the *melanocytic nevi (nv)* class accounts for more than 65% of the data. This imbalance can cause models to become biased toward predicting the dominant class.

***Figure 1: Class Distribution***




*As shown in Figure 1, the dataset distribution is highly skewed, with the nv class significantly dominating other categories. This imbalance directly affects model learning and motivated the use of imbalance mitigation strategies in this project.*

## 4. Methodology

### 4.1 Initial Limitation (Baseline Model)

The initial approach utilized a basic CNN trained for only 5 epochs. Although it achieved an accuracy of approximately 68–69%, further analysis revealed that the model predominantly predicted the majority class.

This issue, known as **majority class collapse**, resulted in near-zero recall for several minority classes. Therefore, the observed accuracy was misleading and did not reflect true model performance.

### 4.2 Improved Experimental Design

To overcome these limitations, the experimental setup was redesigned with a focus on:

- Explicit handling of class imbalance
- Adoption of transfer learning
- Controlled and longer training procedures
- Use of appropriate evaluation metrics

This redesign introduces experimental rigor and enables a more meaningful analysis of model performance.

### 4.3 Handling Class Imbalance

To address class imbalance, multiple strategies were applied:

- **Strategic undersampling:** Reduced dominance of the *nv* class
- **Class weighting:** Increased penalty for misclassifying minority classes
- **Data augmentation:** Improved representation of underrepresented classes

These techniques ensured that the model learns meaningful features across all classes rather than focusing only on the majority class.

### 4.4 Data Augmentation

Data augmentation techniques were applied to improve generalization and reduce overfitting. These include:

- Rotation (±20°)
- Horizontal and vertical flipping
- Width and height shifts

These transformations simulate real-world variations in medical imaging, such as differences in orientation and lighting, enabling the model to learn more robust and invariant features.

### 4.5 Model Architecture (Transfer Learning)

A transfer learning approach was implemented using **MobileNetV2**, pre-trained on ImageNet.

The architecture consists of:

- Global Average Pooling (GAP) to reduce feature dimensions
- Dense layer with ReLU activation for feature learning
- Dropout layer (0.5) to prevent overfitting
- Softmax output layer for seven-class classification

This architecture significantly improves feature extraction compared to a basic CNN.

## 5. Training Strategy

### 5.1 Phase 1 – Feature Extraction

In this phase:

- The base model was frozen
- Only the classification head was trained
- Learning rate was set to 1e-3
- Class weights were applied

Result:

- Validation accuracy ≈ **56%**

This phase stabilizes learning and adapts pre-trained features to the dataset.

### 5.2 Phase 2 – Fine-Tuning

In the second phase:

- The base model was unfrozen
- Learning rate was reduced to 1e-5

- EarlyStopping was applied

Training stopped automatically at Epoch 4 due to no improvement in validation loss.

Best performance:

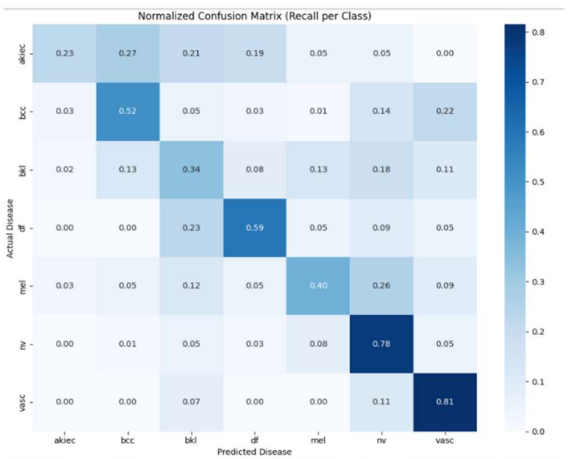
- Validation accuracy ≈ **56.8%**
- Validation loss ≈ **1.13**

This controlled training strategy ensures both generalization and prevention of overfitting.

## 6. Results and Evaluation

### 6.1 Confusion Matrix

***Figure 2: Normalized Confusion Matrix***




*As shown in Figure 2, the confusion matrix exhibits strong diagonal values, indicating correct predictions across multiple classes.*

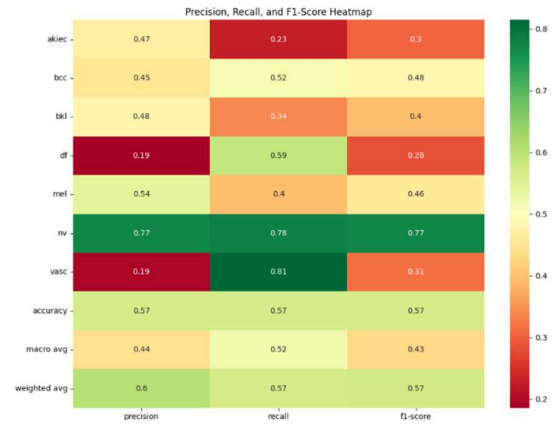
Key observations:

- *vasc*: Recall = **0.81**
- *df*: Recall = **0.59**
- *nv*: Recall = **0.78**

Importantly, unlike the baseline model, no class has near-zero recall. This confirms that the model no longer collapses to predicting only the majority class.

### 6.2 Classification Metrics

***Figure 3: Classification Metrics Heatmap***




Final performance metrics:

- Accuracy: **0.57**
- Macro Recall: **0.52**
- Macro F1-score: **0.43**

Although the overall accuracy is lower than the baseline, these metrics provide a more realistic evaluation by reflecting balanced performance across all classes.

## 7. Critical Analysis

### 7.1 Comparison with Initial Model

| **Aspect** | **Initial Model** | **Improved Model** |
|---|---|---|
| **Accuracy** | 68–69% | 57% |
| **Minority Recall** | 0 | High (e.g., 0.81 for vasc) |
| **Behavior** | Majority bias | Balanced predictions |

The initial model achieved high accuracy by exploiting class imbalance, making it unreliable. In contrast, the improved model provides significantly better class-wise performance.

### 7.2 Key Improvements

- Effective handling of class imbalance

- Adoption of transfer learning architecture
- Implementation of controlled training strategies
- Use of appropriate evaluation metrics
- Improved interpretability through visualizations

**7.3 Interpretation**

The results demonstrate that:

- Accuracy alone is not sufficient for imbalanced datasets
- Macro-based metrics provide a more reliable evaluation
- Proper experimental design leads to meaningful improvements

This demonstrates that the improved model provides a more truthful and clinically relevant assessment of performance.

**8. Conclusion**

This project demonstrates a clear progression from a basic CNN model to an advanced and analytically rigorous deep learning system.

By addressing class imbalance, applying transfer learning, and using appropriate evaluation metrics, the final model achieves balanced performance across all classes. This makes it more suitable for real-world medical applications.

**9. Future Work**

Future improvements can be guided by observed model limitations:

- Applying **focal loss** to further address class imbalance
- Fine-tuning deeper layers of the network for improved feature learning
- Optimizing classification thresholds to improve recall for critical classes such as melanoma (*mel*)
- Exploring advanced architectures such as EfficientNet for improved performance

**10. Summary of Improvements**

This submission directly addresses previous feedback:

1. Class imbalance was handled using weighting, sampling, and augmentation
2. A basic CNN was replaced with a transfer learning approach (MobileNetV2)
3. Training was improved using a two-phase strategy (feature extraction and fine-tuning)
4. Controlled training techniques such as EarlyStopping were applied
5. Evaluation was enhanced using macro-F1 and recall instead of relying only on accuracy
6. Analysis was strengthened through detailed interpretation and explicit figure referencing